MIL-I-81765/4A
15 June 1987
SUPERSEDING
MIL-I-81765/4(AS)
20 JULY 1970

MILITARY SPECIFICATION SHEET

INSULATING COMPONENTS, MOLDED, ELECTRICAL, HEAT SHRINKABLE FLUOROELASTOMER, FLEXIBLE, CROSSLINKED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the insulating components described herein shall consist of this document and the issue in effect of MIL-I-81765.

REQUIREMENTS:

Continuous operating temperature range: -55°C (-67°F) to + 200°C (+392°F)

Color: Unless otherwise specified, molded components fabricated to the requirements herein shall be furnished in black.

Properties: Properties shall be in accordance with Table I.

TABLE I. Physical properties

Property	Requirement	Test Method
Dimensions, inches As received After shrinkage	Shall be in accordance with the applicable drawing, specification or procurement document	4.6.1 Shrinkage - 200°± 5°C (392° ± 9°F) for 10 minutes
Elastic memory, %, min Expansion Retraction	200 90	4.6.2 200°± 5°C (392°± 9°F) for 5 minutes; 4 inch stretch
Tensile strength, psi, min	1800	4.6.3
Ultimate elongation, %, min	300	4.6.3

AMSC N/A FSC 5970

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TABLE I (CONTINUED)

Property	Requirement	Test Method
Low temperature flexibility	No cracking	4.6.4 -55°± 1°C (-67°± 2°F) for 4 hours
Heat shock	No dripping, flowing or cracking	4.6.5 300°± 5°C (572°± 9°F) for 4 hours
Heat resistance Tensile strength, psi, min Elongation, %, min	 1600 250	4.6.6 250°± 3°C (482°± 5°F) for 168 hours
Fluid resistance Tensile strength, psi, min Elongation, %, min	 1600 200	4.6.7 24 hours at 25°± 1°C) (77°± 2°F)
Corrosion	No corrosion	4.6.8 175°±3°C (347°±5°F) for 16 Hours
Hardness, Shore A	85 ± 5	ASTM D 2240
Stiffness, psi, max	7500	ASTM D 747
Dielectric strength, volts/mil, min	200	ASTM D 149
Volume resistivity, ohm-cm, min	10^{10}	ASTM D 257
Water absorption, %, max	0.5	ASTM D 570 24 Hrs @ 23°C (73°F)
Flammability Ave. time of burning, sec., max. Ave. extent of burning, in. (mm) max	 90 1 (25)	ASTM D 635
Fungus resistance	Rating of 0	ASTM G 21
Specific gravity, max	1.95	ASTM D 792

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Shelf life (as supplied): The manufacturer shall certify in writing that the molded components, after storage at a temperature of 18°to 35°C (65°to 95°F) for a period of one year, shall conform to the dimensional requirements specified herein and in the contract or purchase order.

Custodians:
Army - MR
Navy - AS
Air Force - 20

Review Activities:
DLA - GS

Preparing Activity:
Navy - AS
(Project No. 5970-0657-4)

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